

OPEN CLUSTER

Pleiades



CATALOG NUMBER
NGC 1435

DISTANCE FROM SUN
380 light-years

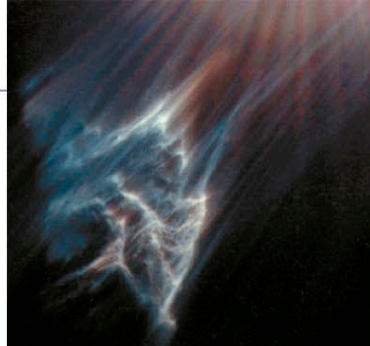
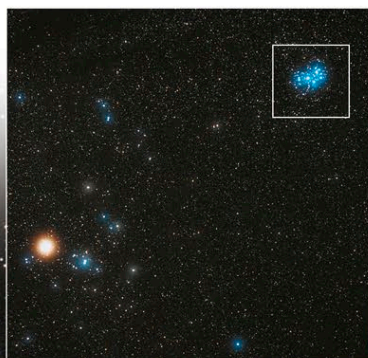
MAGNITUDE 4.17

TAURUS

The Pleiades, also known as the Seven Sisters, is the best-known open cluster in the sky and has been recognized since ancient times (see panel, right). The cluster is easily visible to the naked eye, but although most people

CLUSTERS IN TAURUS

The two best-known clusters, the Pleiades (boxed) and the Hyades (opposite), both lie in Taurus. However, the Pleiades is more than 200 light-years more distant.



GHOSTLY NEBULA

This haunting image shows an interstellar cloud caught in the process of destruction by strong radiation from the star Merope in the Pleiades. The cloud is called IC 349 or Barnard's Merope Nebula.

can see seven stars in the Pleiades, the seventh can often be elusive. Nine stars can be seen on a very dark and clear night. The nine brightest stars are known as the father, Atlas; the mother, Pleione; and the sisters Alcyone, Maia, Asterope, Taygeta, Celaeno, Merope, and Electra. Small telescopes and binoculars reveal many more stars, and larger telescopes show that the cluster, in fact, contains hundreds of stars. The Pleiades is about 100 million years old and will remain a cluster for only

another 250 million years or so, by which time it will have broken up into separate isolated stars. The stars of the Pleiades are blue giants of spectral class B (see pp.232–233) and are hotter and more luminous than the Sun. Long-exposure photography reveals that the Pleiades stars are embedded in clouds of interstellar dust. The clouds are illuminated by radiation from the stars, and they glow as reflection nebulae (see p.228). Although most gas and dust surrounding star clusters represents the material that gave birth to the stars, here the clouds are merely moving through the cluster. The clouds are traveling relative to the Pleiades at 25,000 mph (40,000 kph) and will eventually pass through the cluster and travel into deep space, where they will once again become dark and invisible.

GLOWING NEBULOSITY

The stars of the Pleiades are surrounded by clouds of dusty material that are reflecting the blue light of the stars. However, the stars were not produced from this material, which seems simply to be passing by.

EXPLORING SPACE

BRONZE AGE CLUSTER

The Nebra Disk is perhaps the oldest semirealistic depiction of the night sky. It was discovered in 1999, near the German town of Nebra, and other artifacts found nearby have allowed it to be dated to about 1600 BCE. The disk depicts a crescent moon, a full moon, randomly placed stars, and a star cluster likely to be the Pleiades. Although its authenticity remains uncertain, the Nebra Disk may be proof that European Bronze Age cultures had a more sophisticated appreciation of the night sky than had previously been accepted.



ANCIENT PLEIADES

The cluster of seven gold dots (above and right of center) has been interpreted as the Pleiades as it appeared 3,600 years ago.